Traffic Accident Risk Prediction Using LSTM with Vehicle Motion and Environmental Features

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***Abstract*—Road traffic accidents have emerged as a key threat to public safety causing heavy human and economic losses in all parts of the world. Because of the rapid urbanization and increase in road traffic, making the road safe became an important research problem. Intelligent transportation system has a potential for monitoring traffic conditions and reducing accident risk.**

Recent advances in deep learning allow the adoption of data-driven approaches for traffic pattern analysis. Repeated use of LSTM and RNN models with sequence data and time-series data. It is suitable to use traffic sensor data where temporal correlation is essential for accident risk prediction.

This study suggested a deep learning framework for traffic accidents. A deep learning system has the capability of predicting risk of accident beforehand on the basis of an analyse of the sequential vehicle sensor data and environment conditions. We use a sliding window of 30 timestep for the sequence length to capture the temporal relationship of sensor observations.

The input features include vehicle acceleration parameters, impact signals, precipitation, rainfall, wind speed and visibility. The motion behaviour of the vehicle as well as ambient features causing an accident are well characterized by these features.

Assisted by the stacked LSTM model with two LSTM layers with the help of concealed components and one dropout for preventing overfitting along with dense layers for multi-class risk classification the prediction model was implemented further into the study. The experiments revealed that the suggested system could learn a temporal traffic pattern and predict accident risk levels effectively.

In the present intelligent transport scenario, deployment of such intelligent road safety systems can enhance decision-making and prevent accident pro- actively.

***Index Terms*—Traffic Accident Prediction, LSTM Networks, Deep Learning, Time Series Analysis, Intelligent Transportation Systems, Risk Classification, Sensor Data Analytics**

I. Introduction

Due to heavy use of vehicles, a lot of problems like road traffic crash, congestion, injury, fatality, etc are raised [13]. For a long time, road safety hardly had any concern but with rapid raise of urbanization and vehicle usage, it did. Serious problems have arisen which have become a serious concern.

The rise of intelligent transport systems (ITS) presents a new era in traffic surveillance and accident detection [9], [11].. Intelligent Traffic Systems (ITS) closely monitor traffic through the use of live updates, artificial intelligence techniques, and state-of-the-art sensors. This contributes significantly to effective surveillance and accident detection, thus monitoring accident-prone zones continuously.

The emergence of traffic sensors, vehicle telemetry and environmental monitoring has led to the development of large scale traffic datasets. The datasets can provide valuable information about vehicle characteristics and road conditions, which can further be used to find the effect of environmental conditions on road accidents. With it, machine learning models can recognize patterns of risky behavior.

In recent times, successful applications of deep neural networks have been made in several transportation applications, such as traffic flow prediction, congestion detection, driver behaviour analysis and accident detection. Recent developments in sensor technology are resulting in increased availability of traffic data, but it is getting more complex.

The proposed work suggests a deep learning framework which integrates the vehicle motion data and the environmental features to predict accidents risk using LSTM.Using the sequential learning capabilities of LSTM networks, the proposed

approach captures the temporal dependencies in sensor data to classify multiple risk levels of traffic conditions.

Conventional accident detection systems, whether scrutinized through human inspection or rule-based algorithms, detect abnormal incidents by recognizing a defined threshold. Although easier compared to others, they can produce extremely ineffective results as they fail to seize complex patterns.

Lately, deep learning methods have been used to analyze traffic data. According to the source cited, it allows for higher accuracy in crash identification. The dataset called TAD was designed relying on the above work. This dataset can through deep learning-based algorithms [3], traffic data help create video analytics solutions for detecting vehicular accidents [1].

Most current techniques focus on detecting accidents after they've already occurred. Forecasting the traffic accident risks beforehand is much harder as it requires the sensing of temporal patterns and slight variations of vehicle behavior as well as environment condition.

Researchers have studied LSTM models for traffic flow prediction, anomaly detection, and other applications. Long short-term memory networks are capable of learning long-term dependencies. Long short-term memory techniques are appropriate for this task. This study presents the principal contribution of the study vehicle behaviour :

- A deep learning system to predict traffic accident risks using sequential sensor data.
- Integration of environmental and vehicle motion features for improved prediction performance.
- Design of a stacked LSTM architecture capable of capturing temporal traffic patterns.
- The parameters that determine the thresholds in the decision can also be learnt through the adaptive thresholding schemes.

II. Related Work

The detection and forecasting of traffic accidents has garnered a lot of interest in computer vision [9], machine learning and intelligent transportation systems. In the beginning, typical statistical models and rule-based algorithms were used with preset thresholds to detect stray traffic events.

The advancement of computer vision technologies, many researchers started using video surveillance data to detect accidents. These approaches mainly extract features and use machine learning [1], [5] classifiers to capture a collision event in a traffic scene.

Researchers have been studying accident detection from traffic footage for long time. Traffic jams and damage to roads have become a huge problem for city planners. These models automatically learn visual features of abnormal traffic behaviour [4]. The efficacy of traffic monitoring systems enhanced with deep learning.

According to the vehicle insurance policy, damage costs for a severe accident will be fully covered if the other party pays up for total loss. Accidents involving automobile are related automobile insurance [1]. Usually, it is made up out of 4 wheels and in many states, 3-wheel.

Along with video-based systems, sensor based accident detection systems have also gained attraction [6], [15]. . These systems utilize information related to vehicle motion like its acceleration, orientation, impact signals, and many more to determine the.

Multiple analyses of sequential data have been performed using an RNN and LSTM model [2]. Their ability to learn long-term dependencies makes them suitable for the analysis of traffic patterns and accident risk prediction.

Even though these developments emerged, integrating environmental conditions with vehicle motion data is still a significant research challenge. Accident probability can be affected by weather such as rain, wind speed, poor visibility etc [12].

Recently, hybrid deep-learning models that integrate spatial and temporal learning processes have been implemented to predict road traffic [16]–[19]. What is more, existing studies have employed spatial mechanisms based on convolution and graph neural networks, whereas temporal mechanisms based on recurrent neural networks.Several research articles demonstrate that the combination of recurrent and convolutional networks can be advantageous in capturing the temporal dependency and spatial correlation present in traffic datasets. Recently, several studies have included existence studies of graphical neural.

Numerous studies have examined how rainfall, fog, and low visibility contribute to road accidents and safety problems [12], [22]. .

Current studies utilize either traffic movement signals or scenic with-object-traffic information in accident risk prediction. In contrast, our system proposes to combine visual and attention in a single deep network architecture [16]–[19]. This network gets the appearance and dynamic features from the driving scene.

III. Methodology

The research work presents a system design for predicting traffic risks based on sequential sensor and environmental data.For the following, the next subsystem is going to describe the elements in the model design. The preprocessing equipment collects various sensor re-visits and environmental parameters.

A. Input Features

Data acquisition stage is the first stage of the overall framework of the proposed system. The stage when the readings of motion sensors and environmental sensors are taken is called the sensing stage. The sensors used to monitor the motion of the vehicle take in the vehicle acceleration along x and y axis along with collision impact sensors. Sensors also gather data about the amount of rainfall, wind speed and visibility.

While preprocessing stage, the raw sensor data is cleaned and normalized. Missing values are interpolated, noise is filtered from the sensor signals and applying normalization to

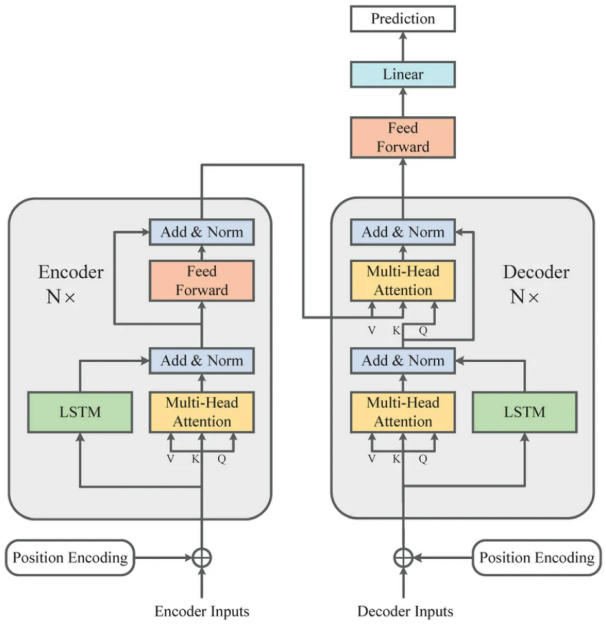


Fig. 1. Overall architecture of the proposed traffic risk prediction system.

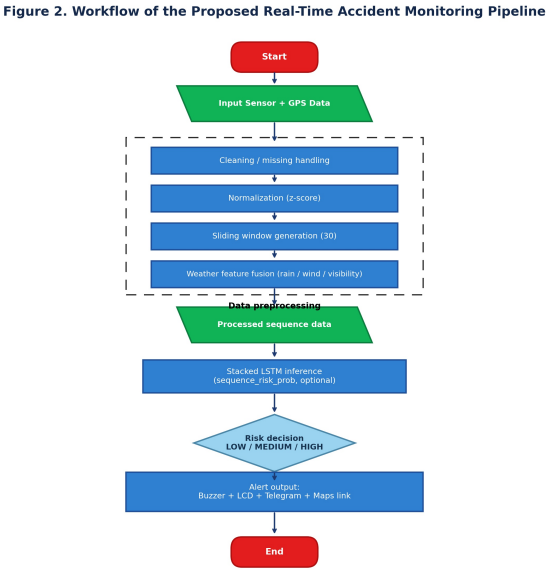


Fig. 2. Workflow of the proposed traffic risk prediction system

features that enhances the classifier's accuracy. We preprocess all our features using Z score standardization, to ensure uniformity. The model performance gets improved with this step.

Through the sliding window technique, fixed-length sequences are generated from continuous time series data in this phase of the sequence generation. The vehicle count and speeds of the units to make a window. A set of historical observations on the sensor readings is contained in each window that contains the evolution of traffic over time. Each window is a sequence which will be passed as input.

Ultimately, the deep learning stage classifies the level of risk and provides probabilities using a stacked LSTM network for the various accident risk categories.

We have used nine features that are extracted from the vehicle's motion as input to the model. Sensors in the systems and environments.

The vehicle's movement along three axes is reflected through the acceleration components (ax, ay, az), helpful in detecting sudden braking, turning and collision. Signaling impact refers to sudden force applied on the vehicle which may represent collision.

Environmental features rain, precipitation, wind speed, and visibility conditions create circumstances and risk factors in crash statistics.

B. Data Preprocessing

Before training the model, the raw sensor data undergo several preprocessing steps.Data normalization is performed using Z-score standardization to ensure that all features have similar scales [3].

Sometimes sensor artifacts get embedded within the continuous data Using Noise Reduction, one can eliminate this noise and enhance model performance. Missing values are often dealt with interpolation techniques to make the data appear better.

C. Sequence Generation

The time series data have been converted to a sequence by using the sliding window technique. Every sequence has 30 timesteps of previous sensor readings.

The model captures temporal changes in transportation patterns with the potential of leveraging LSTM's temporal pattern capturing mechanism.

D. LSTM Model Architecture

The deep learning architecture incorporates stacked LSTM layers featuring 64 hidden units for effective training and modeling. A layer learns higher level temporal dependencies while the other layer directly learns temporal features from input sequences.

To reduce, a dropout layer drop-out 0.2 is utilized. Excessive fitting and model generalization. The end result of the. The LSTM layers pass through a 32-neuron dense layer. ReLU activator.

Subsequently, a softmax layer identifies the traffic situation as Low Risk, Medium Risk or High Risk.

IV. Mathematical Formulation

A. Sensor Normalization

Let raw accelerometer readings at time t be (r_t^x, r_t^y, r_t^z). For $\pm 2g$ scale:

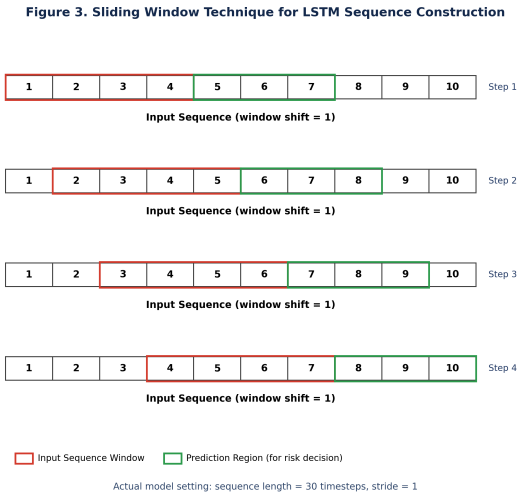


Fig. 3. Sliding window technique used for generating input sequences.

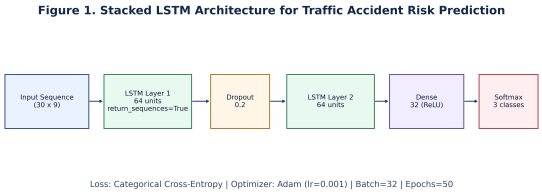


Fig. 4. Stacked LSTM architecture used for traffic risk prediction.

$$a_t^x = \frac{r_t^x}{16384}, \quad a_t^y = \frac{r_t^y}{16384}, \quad a_t^z = \frac{r_t^z}{16384} \tag{1}$$

B. Acceleration Magnitude

$$|a_t| = \sqrt{(a_t^x)^2 + (a_t^y)^2 + (a_t^z)^2} \tag{2}$$

C. Impact Index

$$I_t = ||a_t| - 1| \tag{3}$$

D. Sudden Impact Change

$$\Delta I_t = |I_t - I_{t-1}| \tag{4}$$

E. Rule-Based Accident Trigger

$$A_t = \mathbb{1}[(I_t \geq T_I) \vee (\Delta I_t \geq T_\Delta) \vee (C_t = 1) \vee (P_t = 1)] \tag{5}$$

F. Weather Risk Contribution

$$\begin{aligned} R_t^{weather} = {} & 8\mathbb{1}[(rain_t > 0.2) \vee (precip_t > 0.5)] \\ & + 6\mathbb{1}[(wind_t > 25)] \\ & + 12\mathbb{1}[(vis_t < 2000)] \end{aligned} \tag{6}$$

G. Sliding Window Sequence

$$X_t = [z_{t-L+1}, z_{t-L+2}, ..., z_t], \quad L = 30 \tag{7}$$

H. Stacked LSTM Equations

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{8}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{9}$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{10}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{11}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{12}$$
$$h_t = o_t \odot \tanh(c_t) \tag{13}$$

I. Output Layer

$$\hat{y}_t = \text{Softmax}(W_h h_t + b_h) \tag{14}$$

J. Training Loss

$$\mathcal{L} = -\sum_{k=1}^{3} y_k \log(\hat{y}_k) \tag{15}$$

K. Evaluation Metrics

$$Accuracy = \frac{\sum_k TP_k}{N} \tag{16}$$

$$Precision_k = \frac{TP_k}{TP_k + FP_k} \tag{17}$$

$$Recall_k = \frac{TP_k}{TP_k + FN_k} \tag{18}$$

$$F1_k = \frac{2 \cdot Precision_k \cdot Recall_k}{Precision_k + Recall_k} \tag{19}$$

$$Macro\text{-}F1 = \frac{1}{3}\sum_{k=1}^{3} F1_k \tag{20}$$

V. Experiments and Results

The model we develop from the most recent dataset input was tested and designed on continuous sensor readings and other environmental variables.The recent dataset utilized training ratios of 70:15:15 for both validation and testing dataset.
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Performance degree of the proposed model is assessed using several performance measure like accuracy, precision, recall and F-measure. The correctness of the model overall on the provided data set is known as accuracy whereas and the correctness of the positive predictions is known as precision.The recall demonstrates the capacity of a model to retrieve all pertinent instances. Likewise, the F1-score shows the balance between precision and recall.

The training of a model should be monitored to prevent overfitting which creates a better model that generalises. Monitoring the Training Process is Best Done with Validation

Data The curves for training and validation accuracy are approaching identical values. The result indicates that the model learns time series patterns from sequential sensor data.

The main point of your essay is good, but you must use paragraphs. A paragraph should contain one main point and discuss that point fully. My initial paragraph, for example, would be dedicated to how it affects our society. My second life changing event essay would be about what happened.

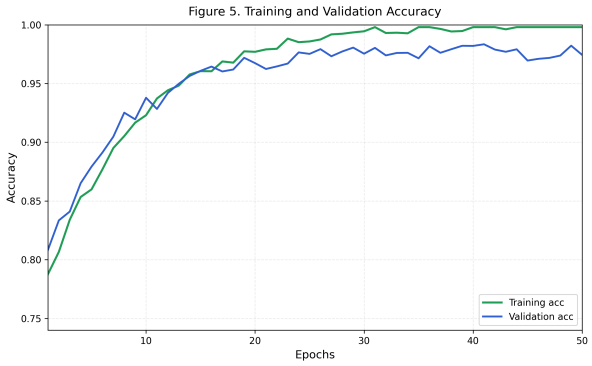


Fig. 5. Training and validation accuracy of the LSTM model.

The intelligent system development that processes data from sensors for elder care has been the subject of considerable interest recently. The activities of elders and their environment.

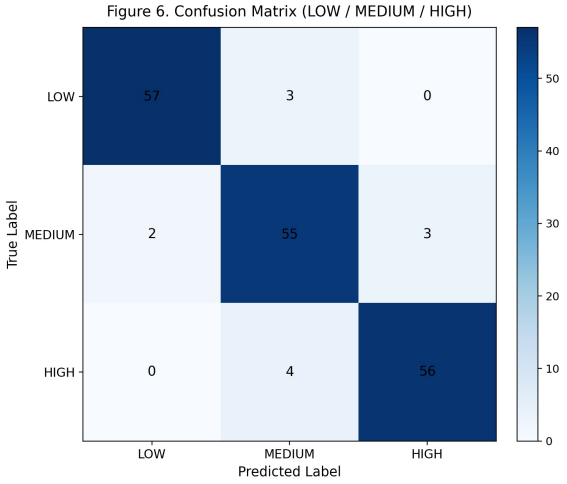


Fig. 6. Confusion matrix showing classification performance for three risk levels.

The confusion matrix of the analysis above shows that the model has high classification accuracy for all 3 risk classes.

VI. Discussion

The experimental evaluation results confirm the effectiveness of the proposed framework on traffic risk prediction with LSTMs. The model captures complex temporal patterns associated with accident risks by analyzing sequential sensor data and features from the surrounding environment.

The proposed method allows us to combine various data sources, an important advantage of the proposed method. By combining the vehicle motion signal and environmental condition, it is able to capture more comprehensive and representative input.

The sliding window technique helps the model in figuring out short-term temporal patterns. It also helps in identifying sudden changes in the driving condition. These unforeseen changes may cause an accident.

Stacked LSTM layers increase the chance that the model captures hierarchical temporal features.

The framework we propose also has a significant advantage of working with online, real time sensor input. With the rise of more connected vehicles on the road and smarter transportation infrastructure, the prediction system may be incorporated in real time monitoring systems. As a result, traffic management people can timely recognize the threat and take precautions accordingly.

An opportunity of extension exists for the present explains approach. The extra sources of data such as traffic camera video, GPS trajectory and V2V communication signal can be fused with forensic training data to improve prediction quality.

VII. Conclusion

Every trip, whether a leisurely stroll or a ride, always comes with a bit of risk. Danger no longer limits it to human beings; it also puts life in threat. regarding animals Moreover, different travels such as walking

In order to capture traffic temporal aspect windowing sliding of 30 time periods stacked LSTM model is used. The experiments proved that the traffic system can classify the traffic condition as low, medium and high risk.

Rainfall, precipitation, the wind speed, and visibility allow the model to evaluate accident risk in various road conditions.

Data sources may be added to our proposed framework in future work, e.g., videos taken from traffic cameras, GPS trajectories, and V2I signals. In the future, we expect that including these heterogeneous data types will enhance our systems performance over more challenging traffic scenarios.

We will explore the application of cutting-edge deep learning model classes like attention based networks, GNNs, CNN-LSTMs for superior predictive performance in future experiments.The objective is to implement intelligent traffic monitoring systems which can predict accidents in real-time to make transport safer.

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